Esteban Alejandro Armas Vega

Personal & Contact Information

Home Address:	Madrid, Spain
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Education

Nov. 2020	PhD in Computer Science, **Complutense University of Madrid** Thesis: “Passive Techniques for Detecting Manipulations in Digital Images”
Oct. 2016	Master of Science in Computer Science, **Complutense University of Madrid** Thesis: “Tools for Dynamic Analysis of Vulnerabilities in Web Applications”
July 2009	Undergraduate Degree in Informatics Engineering **Technological University of Havana “José Antonio Echeverría”** Thesis: “Software Management and Search System ”BuSo””

Scholarships and Certificates

Sept. 2014	2 years Master Scholarship by Secretariat of Higher Education of Ecuador (€50,000)

Languages

Spanish:	Native
English:	C1 Equivalent

Computer Skills

Programming Languages:	Python, Java, C#, JavaScript, Ruby, Swift, PHP, C, C++, SQL, Maude, Prolog
ML Frameworks:	Keras, PyTorch, Tensorflow, Scikit-learn, Caffe
DevOps Tools:	Git, Kubernetes, Docker, Jenkins, Prometheus, Grafana
OS and Cloud Platforms:	macOS, Linux, Windows, GCP, AWS, Azure
Security:	OSINT tools, YARA, Cuckoo Sandbox, MISP, Snort

Personal Skills

Active listening, Ability to take part in technical discussions, Self-starter, Proactive, Problem-solving, Leadership, Management Skills

Interests and Activities

Cybersecurity, Computer Forensics, Malware Analysis, Computer Vision, Machine Learning, Deep Learning, Data Mining, Cryptography

Work Experience

Current – Nov. 2020 | Researcher at ATOS Spain
ATOS ARI Research

As an ATOS' Research and Innovation department (ARI) member, my functions are to research and development frameworks, APIs, and algorithms to enrich cyber threat intelligence using OSINT tools. In addition, I am developing a platform to exchange, in a secure way, sensitive information between financial institutions using the MISP platform as a base. Also, I am in charge of writing deliverables for the European research project tasks in which I am involved and also writing project proposals for Horizon Europe calls.

Current – Dec. 2024 | *Cybersecurity Researcher at* UNIR - *Half Time*
Universidad Internacional de La Rioja

Working half-time as a cybersecurity faculty researcher, I contribute to defining UNIR's cybersecurity research strategy, performing hands-on experimentation, and publishing academic results. I also participate in writing proposals for competitive public/private research grants and in developing a new European cybersecurity academic program. Furthermore, I disseminate research outcomes through conferences and other channels, teach subjects related to Artificial Intelligence, Cybersecurity, and Programming, and supervise final projects. I supervise Master's theses and contribute to grant proposals and curriculum design.

Feb. 2022 – Nov. 2016 | Researcher at Complutense University of Madrid
GASS Cybersecurity Researcher

As a member of the GASS research group of the Complutense University of Madrid, I've been working on research and development of European cybersecurity projects funded by the European Commission (Horizon 2020 & Horizon Europe). Development of computer vision algorithms to detect forgery within Images and Videos. Development of a multimedia forensic analysis platform. Design and development of artificial intelligence algorithms to identify attack patterns in SDN networks. Development of APIs for the early detection of malware and phishing attacks in business networks. As a part of the research, I also write proposals for European research projects and scientific papers for high-impact journals and widely disseminated conferences.

Oct. 2016 – Nov. 2015 | Developer at ESRI Geosystems, Spain
Web and Mobile Applications Developer

Design and development of applications using ESRI software and technology. Design and development of custom applications using open source software and ESRI technology. Research of open-source technologies to integrate them with ESRI software and explore new application areas for its technology. Development of solutions for mobile devices, mainly in iOS. Most of the development was done using the following languages and frameworks: Python, JavaScript, Swift, Objective-C, C#, Spring, MongoDB, Bootstrap. In addition, different tools were used for the development, tools such as GitHub, Maven, JUnit, Subversion, Docker, puppet, etc.

Sept. 2014 – Sept. 2009 | IT Manager at Armas Vega Asociados S.A., Quito - Ecuador
Software Development and Network Management

Development of desktop software applications for inventory, accounting and enterprise's management. Design and deployment of the entire HQ's office data network infrastructure. Configuration and deployment of Servers and Networking Equipment. Design, development and deployment of Web Applications (mainly PHP, JavaScript and MySQL based) for project management.

Research Activities

Publications

M. van Haastrecht, **E. A. Armas Vega**, M. Spruit, **"A Shared Cyber Threat Intelligence Solution for SMEs"**, Electronics. Submission state: Accepted, November. 2021.
doi: doi.org/10.3390/electronics10232913

E. A. Armas Vega, E. Gonzalez Fernández, A. L. Sandoval Orozco and L. J. García Villalba, **"Copy-Move Forgery Detection Technique Based on Discrete Cosine Transform Blocks Features"**, Neural Computing and Applications. October. 2020. doi: doi.org/10.1007/s00521-020-05433-1

E. A. Armas Vega, E. Gonzalez Fernández, A. L. Sandoval Orozco and L. J. García Villalba,**"Image Tampering Detection by Estimating Interpolation Patterns"**, Future Generation Computer Systems.

June 2020. doi: 10.1016/j.future.2020.01.016.

E. A. Armas Vega, E. Gonzalez Fernández, A. L. Sandoval Orozco and L. J. García Villalba, **"Passive Image Forgery Detection Based on the Demosaicing Algorithm and JPEG Compression"**, IEEE Access, 2020. doi: 10.1109/ACCESS.2020.2964516

E. A. Armas Vega, A. L. Sandoval Orozco, L. J. García Villalba: **"Digital Images Authentication Technique Based on DWT, DCT and Local Binary Patterns"**, Sensors, October 1 2018.

A. Lopez Vivar, **E. A. Armas Vega**, A. L. Sandoval Orozco, L. J. García Villalba, T. Kim: **"Ransomware Automatic Data Acquisition Tool"**, IEEE Access, September 1 2018.

F. Román, **E. A. Armas Vega**, L. J. García Villalba: **"Analyzing the traffic of penetration testing tools with an IDS"**, The Journal of Supercomputing, 19 November 2016.

J. Rosales Corripio, **E. A. Armas Vega**, A. L. Sandoval Orozco, L. J. García Villalba: **"Uso de Características en la Identificación de la Fuente de Imágenes de Dispositivos Móviles"**, in Proceedings of the 15th Spanish Meeting on Cryptology and Information Security (RECSI), Maó Menorca, Spain, November 2016.

E. A. Armas Vega, F. Román, L. J. García Villalba: **"Herramientas de Análisis Dinámico de Aplicaciones Web con Snort"**, in Proceedings of the 8th International Congress of Computing and Telecommunications (COMTEL), Lima, Peru, September 2016.

E. A. Armas Vega, A. L. Sandoval Orozco, L. J. García Villalba: **"Poster: Evaluating Black-Box Testing Tools"**, in Proceedings of the 2nd IEEE European Symposium on Security and Privacy, Paris, France, April 26 – 28, 2017.

E. A. Armas Vega, A. L. Sandoval Orozco, L. J. García Villalba, J. Hernandez-Castro, T. Silva, A. Prada: **"Poster: Internet Forensic Platform for Tracking the Money Flow of Financially-Motivated Malware"**, in Proceedings of the 2nd IEEE European Symposium on Security and Privacy, Paris, France, April 26 – 28, 2017.

E. A. Armas Vega, A. L. Sandoval Orozco, L. J. García Villalba: **"Benchmarking of Pentesting Tools"**, in Proceedings of the 19th International Conference on Information Technology, Paris, France, May 18 – 19, 2017.

Participation on International Research Projects

Project: **DIGITAL4Security** – European Masters Programme in Cybersecurity Management and Data Sovereignty

PROPOSAL NUMBER: 101123430.

FINANCING: Digital Europe Programme (DIGITAL)

DETAILS: DIGITAL4Security is a European Master's in cybersecurity management, helping SMEs prevent and respond to threats. It fast-tracks graduates into high-demand roles, bridging the skills gap. The program reskills professionals to enhance security and promotes diversity for a strong cybersecurity workforce. `https://miniurl.cl/DIGITAL4Security`

DURATION: **Since** October 01 2023 **to** September 30 2027

Project: **SAFE** – Security AI for Enhanced SOC
CODE: DIGITAL-ECCC-2023-CYBER-03, Digital Europe Programme, Grant Agreement No. 101190370.
FINANCING: European Union, Digital Europe Programme, supported by the European Cybersecurity Competence Centre (ECCC).
DETAILS: The SAFE project develops AI-enabled technologies to enhance the capabilities of Security Operations Centres (SOCs), focusing on cyber threat intelligence creation and enrichment, automated malware and forensic analysis, risk-based vulnerability management, and AI-assisted incident response. The project aims to improve situational awareness, threat detection, decision-making, and secure cyber threat intelligence sharing across European SOCs. https://ec.europa.eu/info/funding-tenders/opportunities/portal/screen/opportunities/projects-details/43152860/101190370/DIGITAL
DURATION: **Since** January 01 2025 **to** December 31 2027

Project: **CYLCOMED** – Cyber securitY tooLbox for COnnected MEdical Devices
CODE: HORIZON-HLTH-2022-IND-13, Research and Innovation action, Proposal Number: 101095542.
FINANCING: European Commission, Horizon Europe research and innovation Programme.
DETAILS: The EU-funded CYLCOMED project envisages strengthening the cybersecurity of connected, in vitro diagnostic, and software as medical devices, maintaining patient performance and safety and preserving or increasing exchanged private data confidentiality, integrity and availability. https://cordis.europa.eu/project/id/101095542
DURATION: **Since** December 01 2022 **to** November 30 2025

Project: **ELECTRON EU** – rEsilient and seLf-healed EleCTRical pOwer Nanogrid
CODE: SU-DS04-2018-2020, Innovation action, Proposal Number: 101021936.
FINANCING: European Commission, Horizon 2020 research and innovation Programme.
DETAILS: ELECTRON project will address the need to shield against a variety of threats – from cybersecurity incidents and privacy violations to electrical disturbances and severe human errors caused by a lack of relevant training. https://cordis.europa.eu/project/id/101021936
DURATION: **Since** October 01 2021 **to** September 30 2024

Project: **HEROES** – Novel Strategies to Fight Child Sexual Exploitation and Human Trafficking Crimes and Protect their Victims
CODE: H2020-SU-SEC-2020, Research and Innovation action, Proposal Number: 101021801.
FINANCING: European Commission, Horizon 2020 research and innovation Programme.
DETAILS: The HEROES project is structured as a comprehensive solution that encompasses three main components: Prevention, Investigation and Victim Assistance. HEROES will be developed in 36 months by a strong consortium of 27 partners from 17 countries. https://cordis.europa.eu/project/id/101021801
DURATION: **Since** December 01 2021 **to** November 30 2024

Project: **GEIGER** – Geiger Cybersecurity Counter
CODE: H2020-SU-DS-2019, Innovation Action, Proposal Number: 883588.
FINANCING: European Commission, Horizon 2020 research and innovation Programme.
DETAILS: GEIGER is a project proposed by an international group of experts as a response to the cybersecurity challenges of SMEs with a limited background on cybersecurity and a restricted budget. GEIGER will be developed in 30 months by a strong consortium of 18 partners from 9 countries. https://project.cyber-geiger.eu
DURATION: **Since** June 01 2020 **to** November 30 2022

Project: **CyberSANE** – Cyber Security Incident Handling, Warning and Response System for the European Critical Infrastructures
CODE: H2020-SU-ICT-2018, Innovation Action, Proposal Number: 833683.
FINANCING: European Commission, Horizon 2020 research and innovation Programme.
DETAILS: CyberSANE introduces a holistic and privacy-aware approach to security incident management, addressing the complexity of these networks composed of cyber assets hosted on cross-border and heterogeneous Critical Information Infrastructures (CIIs). To meet all of CyberSANE's objectives, the project consortium comprises fifteen of the best organizations in their field from 12 EU countries. https://www.cybersane-project.eu/author/estaddon/
DURATION: **Since** September 01 2019 **to** August 31 2022

Project: **CRYPTACUS** – Cryptanalysis of Ubiquitous Computing Systems
CODE: ICT (Information and Communication Technologies) COST Action IC1403.
FINANCING: COST (European Cooperation in Science and Technology).
DETAILS: Austria, Belgium, Bosnia and Herzegovina, Bulgaria, Croatia, Denmark, Estonia, Finland, France, FYR Macedonia, Germany, Greece, Hungary, Ireland, Israel, Italy, Luxembourg, Netherlands, Norway, Poland, Portugal, Romania, Slovenia, Spain, Sweden, Switzerland, Turkey, United Kingdom
DURATION: **Since** December 12 2014 **to** November 11 2018

Project: **CyberSec4Europe** – Cyber Security Network of Competence Centres for Europe
CODE: H2020-SU-ICT-2018-2, Research and Innovation action, Proposal Number: 830929.
FINANCING: European Commission, Horizon 2020 research and innovation Programme.
DETAILS: With over 100 cybersecurity projects between them, the CyberSec4Europe consortium members cover a wide spectrum of cybersecurity issues: 14 key cybersecurity domain areas, 11 technology/applications elements and nine crucial vertical sectors. https://cybersec4europe.eu
DURATION: **Since** February 01 2019 **to** July 31 2022

Project: **CONCORDIA** – Cyber security cOmpeteNce fOr Research anD Innovation
CODE: H2020-SU-ICT-2018-2, Research and Innovation action, Proposal Number: 830927.
FINANCING: European Commission, Horizon 2020 research and innovation Programme.
DETAILS: CONCORDIA Consortium includes 51 of the most highly qualified EU universities, industries, public bodies and organizations from more than 20 countries around the world. https://www.concordia-h2020.eu
DURATION: **Since** January 01 2019 **to** December 31 2022

Project: **RAMSES** – Internet Forensic Platform for Tracking the Money Flow of Financially-Motivated Malware
CODE: H2020-FCT-2015, Innovation Action, Proposal Number: 700326.
FINANCING: European Commission, Horizon 2020 research and innovation Programme.
DETAILS: Treelogic Telematica y Logica Racional para la Empresa Europea S.L. (España), Policia Judiciaria (Portugal), University of Kent (UK), Research Centre on Security and Crime (Italy), Universidad Complutense de Madrid (Spain), College of the Bavarian Police (Germany), Trilateral Research and Consulting (UK), Politecnico di Milano (Italy), Belgian Federal Police (Belgium), Saarland University (Germany), Spanish National Police (Spain). https://ramses2020.eu
DURATION: **Since** September 01 2016 **to** September 01 2019

Project: **SELFNET** – Framework for Self-Organized Network Management in Virtualized and Software Defined Networks
CODE: H2020-ICT-2014-2, Innovation & Research Action (RIA), Proposal Number: 671672.
FINANCING: European Commission, Horizon 2020 research and innovation programme.
DETAILS: EURESCOM - European Institute for Research and Strategic Studies in Telecommunications (Germany), InnoRoute (Germany), The German Research Center for Artificial Intelligence (Germany), Universidad Complutense de Madrid (Spain), Universidad de Murcia (Spain), Creative Systems Engineering Monoprosopi EPE (Greece), Alvarion Technologies Ltd. (Israel), Nextworks Srl (Italy), PT Inovação e Sistemas (Portugal), Proef (Portugal), Ubiwhere Lda. (Portugal), University of the West of Scotland (UK). https://selfnet-5g.eu
DURATION: **Since** July 1 2015 **to** June 30 2018